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SUMMARY

Machine learning engineer and full-stack developer pursuing B.Tech in Electronics & Computer Engineering at Manipal Institute of Technology (2027). Experienced in end-to-end deep learning model development, computer vision, large language model (LLM) training from scratch, edge AI deployment, and scalable software engineering. Built a 62M-parameter domain-specific transformer that outperforms CodeLLaMA-7B (80× larger) on embedded code tasks. Deployed real-time YOLOv7 segmentation at 22 FPS on NVIDIA Jetson at ISRO. President, MIT Bengaluru Open Source Community (MBOSC).

EDUCATION

Manipal Institute of Technology

Bengaluru, India

B.Tech in Electronics & Computer Engineering | GPA: In Progress

Aug. 2023 – Jul. 2027

- **Leadership:** President, Manipal Bengaluru Open Source Community (MBOSC) – leading 200+ member developer community
- **Coursework:** Computer Architecture, Data Structures & Algorithms, Network Protocols, Embedded Systems, Probability & Statistics

EXPERIENCE

Computer Vision Intern

Dec. 2025 – Jan. 2026

ISRO – Laboratory for Electro-Optics Systems (LEOS)

Bengaluru, India

- Solved the critical problem of orbital data scarcity for non-cooperative satellite target detection by architecting a procedural synthetic data pipeline using BlenderProc, generating **6,000+ high-fidelity annotated images** with automated COCO/RLE labels.
- Improved training data diversity by implementing Uniform SO(3) rotations and spherical camera trajectories, guaranteeing **360-degree angular coverage** and eliminating blind spots in the deep learning model.
- Reduced synthetic data generation time by **70%** by benchmarking path-tracing parameters across a 5×5 configuration matrix, while preserving photorealistic feature fidelity required for model training.
- Deployed a YOLOv7-Segmentation model on NVIDIA Jetson Nano via TensorRT optimisation, achieving **22 FPS real-time inference** on a physical satellite miniature – production-ready edge AI deployment.
- Designed a next-generation simulation roadmap integrating Unreal Engine 5, NASA SPICE toolkit (Max-Q), and GAN-based domain adaptation to close the Sim-to-Real gap for future orbital missions.

Software Engineering Intern

May 2025 – Jul. 2025

SenseOps Tech Solutions Pvt. Ltd.

Bengaluru, India

- Rebuilt and optimised the company's production website with a mobile-first responsive design, improving Google Lighthouse performance scores and reducing page load times by a measurable margin.
- Designed and implemented a **custom UDP/TCP network protocol** for a distributed vibration sensor network, improving real-time data transmission reliability and reducing packet loss across 10+ hardware nodes.

Founder & Lead Engineer

Jan. 2025 – Present

Webcraft Studios

Bengaluru, India

- Founded a software and digital agency from scratch; scaled to **multiple international SaaS clients** including Myntra Minis (Flipkart ecosystem) and Str8Fire.io within the first year of operation.
- Engineered full-stack web platforms using React, Next.js, Node.js, and PostgreSQL with Stripe-integrated subscription billing, user authentication, and real-time dashboard analytics.
- Built a structured client delivery pipeline (Discovery → Design → Development → Launch → Support) that standardised project execution and reduced average delivery cycle time by over **30%**.

PROJECTS

EmbedGen – Domain-Specific Code LLM | *Python, PyTorch, ONNX, FastAPI, TypeScript*

2025

- Designed and trained a **62M-parameter decoder-only transformer** from scratch using a Llama-lite architecture with RMSNorm, Rotary Positional Encoding (RoPE), and SwiGLU activations, trained entirely on embedded systems code (STM32 HAL, Arduino, ESP32, FreeRTOS).

- Built the complete data pipeline: custom GitHub/Arduino scrapers, BPE tokenizer with 32k vocabulary, MinHash LSH deduplication, and quality filtering to produce a **curated 500k-pair instruction dataset** – demonstrating that domain-specific pretraining beats model scale.
- Created **EmbedEval**, a domain-specific evaluation benchmark of 50 embedded programming problems across 10 categories (GPIO, UART, DMA, FreeRTOS, ISR, I2C, SPI, ADC, Timers, Interrupts), with correctness validated by arm-none-eabi-gcc compilation.
- **EmbedGen outperforms CodeLLaMA-7B** – a model 80× larger – on EmbedEval, validating the hypothesis that narrow domain training is more effective than general-purpose scaling for specialised code generation.
- Optimised and deployed the model via ONNX export with FP16 quantisation on a FastAPI local inference server, achieving fully offline operation; shipped a **VS Code extension** with ghost-text autocomplete for C/C++ embedded development.

Bird Species Audio Classifier | *Python, TensorFlow, Keras, Librosa, CNNs*

2023

- Designed and trained a convolutional neural network (CNN) on mel-spectrogram representations of raw bird audio clips, achieving **90%+ classification accuracy** across multiple species – demonstrating applied signal processing and deep learning for audio classification.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C, C++, HTML/CSS, SQL

AI / ML & Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, YOLOv7, TensorRT, ONNX, BlenderProc, Librosa

NLP / LLM: Transformer Architecture, Tokenization (BPE), RoPE, RMSNorm, SwiGLU, ONNX Quantisation, Instruction Tuning, Model Evaluation

Web & Full-Stack: React, Next.js, Node.js, React Native, REST APIs, Tailwind CSS, MongoDB, PostgreSQL, Firebase, Stripe API

Tools & Infrastructure: NVIDIA Jetson, Docker, AWS, Vercel, Git, GitHub, FastAPI, Figma, Unreal Engine 5

Concepts: Machine Learning, Deep Learning, Computer Vision, Edge AI Deployment, Synthetic Data Generation, Sim-to-Real Transfer, Embedded Systems, Agile Development